

# Anurag Pandey

Email: [Anurag.120453@stu.upes.ac.in](mailto:Anurag.120453@stu.upes.ac.in) | GitHub: [AnuragPandey19](https://github.com/AnuragPandey19) | LinkedIn: [anurag-pandey-154259280/](https://www.linkedin.com/in/anurag-pandey-154259280/) | +91 9580765051 | India

## Education

UPES, Dehradun

B.Tech C.S.E AI/ML

Aug 2023 – May 2027

7.7 CGPA

## Professional Summary

Software Engineer and Machine Learning Engineer with hands-on experience building end-to-end applications — from training and deploying ensemble ML models to architecting REST APIs, databases, and browser extensions. Comfortable across Python, Java, and JavaScript with exposure to both full-stack development and production ML pipelines.

## Professional Experience

IBM

June 2026 – July 2026

AI/ML Intern

Live: [undebuggedbit-chimera-fd.hf.space](https://undebuggedbit-chimera-fd.hf.space)

- Built and deployed CHIMERA-FD, a live cascaded fraud-detection platform (LightGBM + isotonic recalibration) reaching 92.13% accuracy with zero engine errors on 3M held-out rows generated by a 12-agent test-case pipeline.
- Architected the multi-tenant B2B stack — FastAPI, Next.js 15, Postgres, JWT/bcrypt auth, Dockerized CI/CD on Hugging Face Spaces — with tenant isolation enforced by an integration test suite.

Notebrain Software & Services Pvt. Ltd

Aug 2025 – October 2025

AI/ML Intern

- Owned the end-to-end ML pipeline for a sports-outcome prediction feature in a live web app: data cleaning, feature engineering, training, and validation.
- Fixed dataset annotation and class imbalance across multiple sources, then tuned models iteratively in an agile team to lift prediction accuracy.

Venture Launcher

June 2025 – July 2025

AI/ML Intern

- Performed data analysis and preprocessing for an investor matching system, including data cleaning, feature identification, and exploratory analysis.
- Built a prototype ML model for matching relevance as a proof of concept for the founding team.

## Projects

SurfVigil — Real-Time Browser Threat Detection Extension | [Extension link](#)

- Shipped a Chrome extension + FastAPI backend running a 4-model threat ensemble (phishing, XSS, cryptomining, DOM anomaly); deployed on Hugging Face Spaces + Neon Postgres and published to the Chrome Web Store.
- Debugged three silent model failures returning false-clean results, then recalibrated ensemble thresholds to cut the false-positive rate from ~65% to ~0% with no loss in genuine detections.

Conversational Assessment Recommender (Agentic RAG) [Live Demo](#)

- Built a multi-turn agentic-RAG recommender over a 377-item catalog, using a hybrid BM25 + MiniLM retrieval pipeline with a URL-grounding guardrail that eliminates hallucinated recommendations.
- Shipped it as a stateless, provider-swappable (Groq/Gemini) FastAPI service on Hugging Face Spaces, with an eval harness scoring schema compliance and Recall@10.

Smart City Guide — Java Swing Desktop Application | [Releases V1.0](#)

- Built a 4-layer Java Swing app over a 7-table MySQL schema (JDBC), with PBKDF2-HMAC-SHA256 auth and prepared-statement queries closing the SQL-injection surface.
- Designed a self-bootstrapping first-run flow and packaged v1.0 as a Windows installer with bundled JRE via jpackage — zero manual setup.

Rice Leaf Disease Detection System | [Website](#)

- Built a two-stage, 5-model ensemble (EfficientNet-B3/DenseNet-121/MobileNetV3 → ViT-Base/ConvNeXt-Tiny) hitting 96.68% triage and 98.18% refinement accuracy; trimmed 6→5 models to cut inference time 16% while gaining 0.91% accuracy.
- Deployed end-to-end via PyTorch + Flask on Hugging Face Spaces with a REST API, Docker, and sub-2s inference.

## Technical Skills

Languages: Python, Java, JavaScript, TypeScript, HTML, SQL, CSS

ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM, SHAP, Pandas, NumPy

GenAI/LLM: RAG, Hybrid Retrieval (BM25 + Dense Embeddings), Sentence-Transformers, LLM Agents, Prompt Engineering

Backend & Databases: FastAPI, Flask, SQLAlchemy, REST APIs, JWT / bcrypt auth, PostgreSQL, MongoDB

Tools & Platforms: Git/GitHub, Docker, Hugging Face Spaces, GitHub Actions (CI/CD), Chrome Extension APIs (Manifest V3)

Core Concepts: Data Structures, Algorithms, OOP, Version Control, Model Deployment, Feature Engineering, Model Calibration, Multi-tenancy

**Position of Responsibility**

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**Core Member – UPES ACM & ACM-W Student Chapters**

- Served as Program Head for *ICPC Code Nexus*, managing logistics and tech ops.
- Awarded Certificate of Appreciation for excellent execution and contribution.

**Social Outreach Intern — Shri Brahma Ji Siksha Avam Samaj Kalyan Samiti (May 2024 – July 2024)**

- Coordinated education and environment awareness drives across 30+ communities, conducting 150+ field surveys.